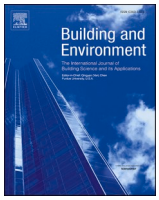
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Interactive effects of noise and odor on construction workers' environmental perception and cognitive performance amid outdoor thermal stress

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ABSTRACT

Construction workers chronically exposed to dynamic outdoor environments constitute a critical cohort for investigating multisensory environmental stress responses and cognitive ergonomics. This study systematically deconstructs the interactive effects of thermal stress gradients and noise–olfactory multimodal stimuli on workers' subjective thermal perception and neurocognitive performance in a cold region of China. Using a controlled field experiment with synchronized micrometeorological monitoring, we simultaneously acquired subjective environmental evaluations, multidimensional cognitive performance metrics, and electroencephalogram (EEG) data from 72 construction workers under nine factorial combinations of sound pressure levels (50/70/85 dBA) and odor types (*Mentha piperita, Citrus sinensis*, and *Rosmarinus officinalis*). A multidimensional benefit assessment model quantitatively ranked intervention efficacy. The results indicate that: (1) noise significantly modulated thermal sensation votes ($p < 0.05$), while thermal stress drove auditory sensation votes ($p < 0.01$). Neurophysiologically, escalating thermal stress enhanced low frequency (δ) activity and suppressed high-frequency (α, β) oscillations, with noise further diminishing frontal α-wave relative power ($p < 0.05$). (2) *C. sinensis* elevated significantly frontal-occipital θ-wave power ($p < 0.05$). (3) *M. piperita* broadly potentiated β-band activity under "neutral" conditions, yet *R. officinalis* yielded significantly lower thermal comfort votes than *C. sinensis* under "warm" conditions ($p = 0.02$). Our findings offer actionable climate adaptation frameworks for occupational health, safeguarding vulnerable workers against intensifying extremes in outdoor working environments.

1. Introduction

1.1. Background

Climate change intensifies heatwave frequency and spatiotemporal severity, escalating heat exposure risks for outdoor workers [1,2]. Prolonged high-temperature exposure induces multisystem physiological stress, psychological overload, work performance decrement, and elevated incidence of life-threatening heat-related illnesses [3,4]. Construction workers represent a particularly vulnerable cohort to heat-induced health impairments and productivity degradation [5,6]. Construction sites intrinsically generate compounded thermal and mechanical noise stressors, with heavy machinery operation persistently exceeding occupational noise exposure thresholds [7]. Chronic noise exposure causes auditory damage, communication barriers, cardiovascular dysfunction, sleep disorders, and neurocognitive decline, while prolonging information processing and response times [8–12]. Critically, these stressors co-occur synergistically, depleting attentional resources, compromising safety protocol adherence, and elevating occupational accident rates [13–16].

Conventional engineering interventions (misting fans, shading, barriers) offer limited mitigation due to high costs, construction disruption, and poor spatial flexibility [17,18]. Olfactory stimulation emerges as a low-cost, minimally disruptive strategy that integrates seamlessly with personal protective equipment [14]. Plant-derived odorants modulate autonomic function via limbic activation, alleviating psychological tension, improving cardiovascular and thermophysiological indices, and enhancing executive function [19,20]. However, extant research remains confined to single sensory pathways or controlled indoor settings,

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Nomenclature

T_a	Air temperature (°C)
RH	Relative humidity (%)
V_a	Wind speed (m/s)
T_g	Global temperature (°C)
T_{mrt}	Mean radiant temperature (°C)
SVF	Sky view factor
OS	Open square
PS	Pavilion space
LAeq	A-weighted equivalent continuous sound pressure level
PET	Physiological equivalent temperature (°C)
MTSV	Mean thermal sensation vote
TSV	Thermal sensation vote
TCV	Thermal comfort vote
ASV	Auditory sensation vote
ACV	Auditory comfort vote
OPV	Olfactory pleasantness vote
OCV	Olfactory comfort vote
EEG	Electroencephalogram
PSD	Power spectral density
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
UTCI	Universal Thermal Climate Index (°C)
TSI	Tropical Summer Index (°C)
WBGT	Wet Bulb Globe Temperature (°C)
HRV	Heart rate variability

lacking systematic evidence on cross-modal regulation of thermal perception and cognition under complex outdoor stressors. Elucidating these mechanisms in construction workers and establishing thermally-calibrated regulatory frameworks is critical for advancing UN Sustainable Development Goals for health (SDG 3) and climate action (SDG 13).

1.2. Literature review

Prior research has established the independent effects of thermal stress, noise, and olfactory stimuli on cognitive function, yet neurobehavioral mechanisms underlying multisensory interactions remain inadequately elucidated. Regarding thermal exposure, field studies at South China construction sites identify thermal acceptability thresholds when physiological equivalent temperature (PET) surpasses 47.0 °C, the Universal Thermal Climate Index (UTCI) exceeds 46.7 °C, or the Wet Bulb Globe Temperature (WBGT) exceeds 34.6 °C [21], with air temperature (T_a) and mean radiant temperature (T_{mrt}) negatively correlating with cognitive task accuracy [22]. Climate chamber experiments reveal neurophysiological substrates: elevated radiant temperatures potentiate β and α oscillations, enhancing alertness and frontal EEG asymmetry [23]. Furthermore, increased Total Heat Stress Index (TCSI) exhibits an inverted U-shaped relationship with cognitive accuracy and a U-shaped relationship with reaction time, indicating that compensatory responses under moderate heat stress transition to functional decompensation at extreme levels [24,25].

A substantial body of literature corroborates the detrimental impact of occupational noise exposure on cognitive performance. Lee et al. [26]. ascertained that complex noise generated by crushers, pile drivers, and hammer compactors elicits the most pronounced detrimental effects on workers' cognitive states, with noise-sensitive individuals exhibiting significantly attenuated work performance. Zeydabadi et al. [12]. further demonstrated that chronic exposure to noise levels exceeding 85 dBA in the metal industry substantially compromises reaction time, attention, and memory relative to unexposed controls. Extending these cognitive findings to behavioral outcomes, Yang et al. [27]. documented significant inverse correlations between construction noise and safety compliance behaviors among 375 workers. At the neurophysiological level, Ke et al. [28]. employed electroencephalographic monitoring to identify elevated β- and γ-band power at T7 and AF4 channels as objective biomarkers of sustained attention and attentional failure in construction personnel.

As the most direct affective conduit among sensory modalities, olfaction circumvents thalamic relay stations to directly engage limbic structures, thereby mediating immediate neuromodulatory control over emotional arousal and stress perception [29]. Afferent olfactory signals transmitted from the olfactory bulb to the amygdala–hippocampal complex and orbitofrontal cortex can trigger affective activation or anxiolytic responses within sub-second timescales [30]. Accumulating evidence demonstrates that essential oil-derived hedonic odorants confer multifaceted psychological benefits, encompassing positive affect enhancement [31,32], vigilance augmentation [33], and cognitive performance optimization [34,35], substantiating their therapeutic applicability across diverse occupational and clinical contexts [36,37].

In occupational settings, Nasiri et al. [33]. demonstrated that rosemary essential oil aromatherapy significantly enhanced alertness and mitigated sleepiness among shift-working nurses without eliciting adverse reactions. Wang et al. [35]. further reported that orange essential oil exposure significantly augmented parasympathetic-mediated heart rate variability during rest periods compared to odor-free control conditions. Moss et al. [38]. observed differential psychophysiological effects across odorant types: peppermint aroma facilitated memory consolidation and subjective alertness, whereas ylang-ylang induced psychomotor slowing, reduced arousal, and enhanced tranquillity. At the neurophysiological level, Hu et al. [39]. demonstrated that mint odor stimulation elicited distinct event-related potentials at parietal midline (Pz) electrode sites and significantly elevated relative α-band power among preference-grouped participants, indicating enhanced pleasant affect mediated by convergent olfactory–trigeminal activation.

Extant literature has predominantly scrutinized physiological and psychological responses to unimodal environmental stimuli under steady-state indoor conditions. In contrast, real-world outdoor environments engender concurrent exposure to polymodal sensory stressors. Nitidara et al. [40]. simulated multisensory interactions among auditory, thermal, and visual factors in public open spaces, demonstrating that auditory perception intensifies concomitantly with elevated thermal sensation and noise levels. Yang et al. [41]. examined combined thermal and acoustic stress in underground workspaces, revealing that thermal comfort votes (TCV) demonstrate increasing susceptibility to noise perturbation at elevated temperatures, whereas temperature effects on the low-frequency to high-frequency heart rate variability ratio (LF/HF) are amplified under high-noise conditions. Chang et al. [42]. demonstrated that olfactory comfort enhancement via aromatic stimuli mitigates thermal discomfort under high-temperature conditions. Jia et al. [43]. systematically investigated odor–heat composite environments, establishing that olfactory conditions exert superior influence over psychological perception relative to thermal parameters, whereas thermal environments exert negligible effects on olfactory comfort.

Nevertheless, these investigations remain largely constrained to steady-state indoor contexts or unisensory experimental paradigms. Construction workers, as an occupationally distinct cohort subjected to chronic exposure to complex, dynamic outdoor environments, may manifest psychophysiological adaptations that fundamentally reconfigure multisensory stress response patterns, thereby constraining the generalizability of findings derived from general population samples.

Consequently, an exigent imperative exists to establish a multimodal neurobehavioral assessment framework specific to environmental stress in construction occupations, and to elucidate the regulatory mechanisms underpinning olfactory interventions within complex outdoor stress scenarios.

1.3. Research gap

Despite cumulative evidence demonstrating both discrete and synergistic effects of thermal, acoustic, and olfactory stimuli on perceptuocognitive systems, two fundamental constraints limit the ecological validity and translational applicability of extant findings. First, prior investigations have predominantly utilized indoor, climatically controlled settings with steady-state parameters or adopted unisensory experimental paradigms, thereby precluding systematic elucidation of the integrative mechanisms governing multimodal stressor interactions in naturalistic outdoor environments. Critically, such studies systematically neglect the modulatory influence of spatiotemporal dynamics in meteorological conditions and soundscape complexity on neurobehavioral outcomes. Second, the extrapolation of findings derived from general populations to construction workers—an occupational cohort subjected to chronic, high-magnitude environmental stress exposure—lacks empirical validation. Prolonged occupational adaptation likely induces distinct neurophysiological plasticity, manifested as altered stress adaptation thresholds and compensatory allostatic mechanisms, resulting in divergent multisensory integration profiles and olfactory responsivity patterns relative to normative benchmarks.

1.4. Research objective

This study conducted field investigations at an urban plaza and pavilion microenvironment in Xi'an, a typical city in China's cold region, to characterize the neurobehavioral modulatory efficacy of controlled multimodal interventions. Through concurrent acquisition of high-resolution microclimatic parameters, subjective environmental appraisals, multidimensional cognitive performance metrics (response latency and accuracy), and scalp electroencephalographic recordings, we examined the differential impacts of three sound pressure levels (50/70/85 dBA) crossed with three phytogenic olfactory stimuli (*Mentha piperita, Citrus sinensis*, and *Rosmarinus officinalis*) on construction workers' neurocognitive responses across varying thermal stress gradients. The primary objectives of our study were: 1) to decipher cross-modal integration patterns wherein olfactory stimulation modulates subjective thermal and auditory perceptions under combined thermal–acoustic stress; 2) to ascertain odor-specific differential impacts on sustained attention and executive function–mediated cognitive performance; 3) to decode the topographical distribution of spectral power (δ, θ, α, and β bands) across frontoparietal, temporal, and occipital cortices to identify neurophysiological signatures of olfactory-induced arousal; and 4) to establish a multi-criteria decision-analysis framework utilizing entropy-weighted TOPSIS methodology for Pareto-optimal prioritization of multi-objective environmental intervention strategies.

Our findings furnish an evidence-based technical framework for context-sensitive occupational health surveillance systems in intelligent construction ecosystems, the scalable implementation of phytogenic olfactory interventions, and the innovation of human-centric occupational management protocols. These contributions advance theoretical understanding of multisensory stress physiology while offering translational applications for safeguarding occupational health and optimizing workforce productivity in high-risk construction environments.

2. Methods

2.1. Study sites

Xi'an is located at the boundary between the humid subtropical climate (Cwa) and the tropical and subtropical steppe climate (BSk) according to the Köppen climate classification. It is characterized as a warm temperate, semi-humid continental monsoon climate, with hot and humid summers. The highest monthly mean air temperature (T_a) occurs in July (28.08 °C), with a recorded maximum T_a of 39.33 °C. The lowest monthly mean T_a occurs in January (0.62 °C), and the minimum T_a reaches −8.01 °C. The annual average relative humidity (*RH*) ranges from 51.28% to 73.68% [44]. The experiments were conducted in an open square (OS) and a pavilion space (PS) in Xi'an, China (108°5′18″ E, 34°15′44″ N). Unshaded open spaces and building-shaded areas represent common working and resting environments for construction workers during summer. The sky view factor (SVF) represents the proportion of visible sky at a given location; a higher SVF indicates lower spatial coverage and has a significant influence on the outdoor thermal environment (Fig. 1).

2.2. Experimental design

2.2.1. Environmental setup

Acoustic stimulation comprised digitized pile driver emissions (source: https://www.freesfx.co.uk/soundeffects), selected as representative of the complex, high-amplitude soundscapes characteristic of active construction sites, wherein pile driving operations constitute a prevalent and physiologically stressful noise source [26]. To ensure ecological validity while maintaining experimental control within the campus microenvironment, exposure levels were standardized using the A-weighted equivalent continuous sound pressure level (LAeq). Three acoustic intensities were established based on established occupational exposure limits: 50 dBA (ambient reference), 70 dBA (moderate occupational exposure), and 85 dBA (high-risk occupational threshold) [45,46].

Sound pressure levels were quantified using an AWA5688 multifunction sound level meter, calibrated according to standardized protocols [47]. Stimuli were delivered via high-fidelity loudspeaker arrays, maintained at target intensities of 50 ± 3 dB, 70 ± 3 dB, and 85 ± 3 dB. To mitigate extraneous acoustic interference, experimental protocols were temporally scheduled during periods of minimal ambient activity, with strict administrative coordination to prohibit concurrent high-amplitude events and restrict proximal pedestrian traffic within the immediate experimental perimeter, thereby ensuring conformance with preset acoustic exposure parameters.

The olfactory interventions comprised *M. piperita, C. sinensis*, and *R. officinalis*—phytogenic odorants selected based on established psychophysiological efficacy (enhanced alertness, cognitive facilitation, and affective modulation), favorable safety and hedonic profiles, and operational feasibility for integration with personal protective equipment or site-level ventilation infrastructure, enabling low-cost, non-disruptive occupational deployment (Table 1). These volatile botanical concentrates, composed primarily of terpenoids, phenylpropanoids, and bioactive volatile organic compounds [48], were administered via controlled aerosol delivery. A 15% (v/v) emulsion was prepared by diluting 15 ml pure essential oil in 100 ml deionized water, and atomized using a 500 ml ultrasonic diffuser with continuous nebulization [49].

To ensure consistent odorant delivery while maintaining ecological validity for outdoor occupational contexts, experimental trials were restricted to meteorologically quiescent, clear-sky conditions to minimize advective dispersion. The diffusion apparatus (dimensions: 100 × 170 × 250 mm) was positioned within a ventilated enclosure, with the aerosol outlet oriented 20 cm from the participant's nasal breathing zone to standardize exposure concentration. This configuration facilitated controlled, reproducible olfactory stimulation appropriate for assessing neurobehavioral responses in construction environments.

Micrometeorological parameters were quantified at each sampling location using tripod-mounted instrumentation positioned at 1.1 m above ground level. Data were acquired at 1-min intervals encompassing

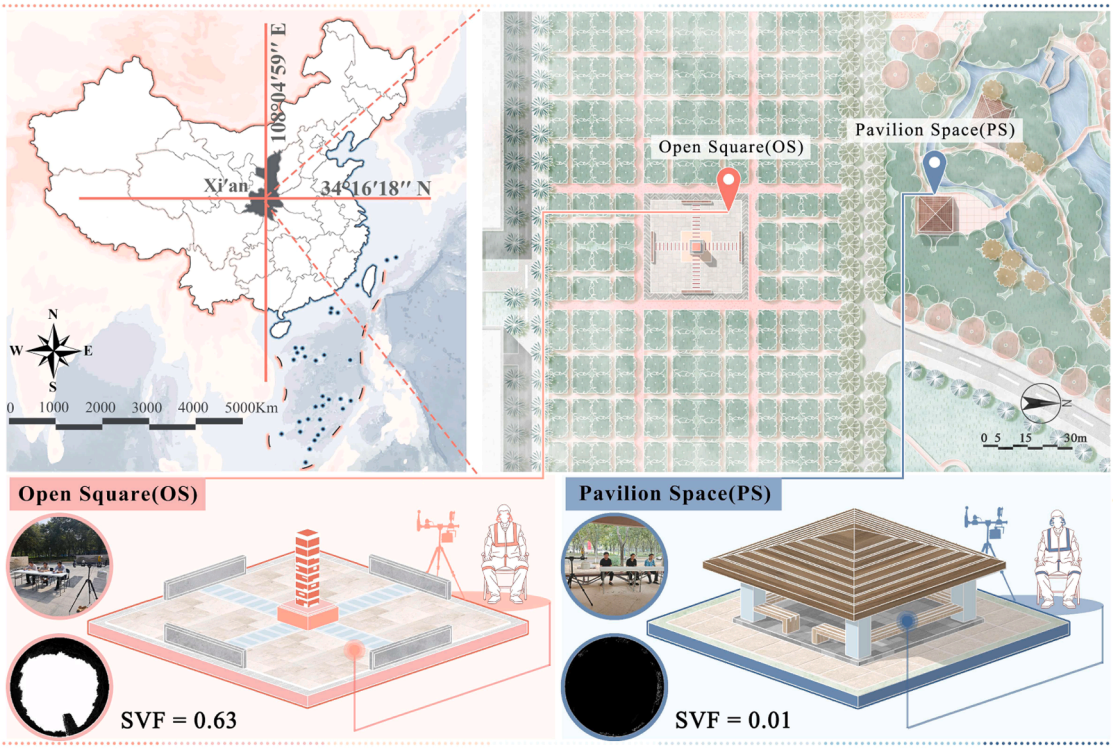


Fig. 1. The study area and measured sites.

T_a, *RH*, globe temperature (T_g), and wind speed (V_a). T_a and *RH* were recorded using a HOBO onset U23–001 (ONSET, USA) data logger housed within a radiation shield. T_g was measured by a black globe thermometer (Delta OHM HD2107.2, Italy). V_a data were collected using a portable weather station (Kestrel 5500, Nielsen-Kellerman Co., USA) (*Appendix Table 1*). The T_{mrt} was calculated from T_a, T_g, and V_a. as follows (*Appendix Table 2*):

$$T_{mrt} = \left[(T_g + 273)^4 + \frac{1.10 \times 10^8 V_a^{0.6}}{\varepsilon D^{0.4}} (T_g - T_a) \right]^{\frac{1}{4}} - 273 \quad (1)$$

where D is the globe diameter ($D = 0.05$ m in this study) and ε is the emissivity ($\varepsilon = 0.95$ for a black globe).

2.2.2. Experimental procedure

Experimental sessions were conducted across six consecutive days of meteorologically stable summer conditions. To ensure sample homogeneity and mitigate confounding variance, stringent inclusion criteria were applied: i) local residency exceeding one year to ensure regional climatic acclimatization; ii) minimum one-year tenure in construction occupations; and iii) absence of diagnosed psychiatric or neurological comorbidities. These parameters controlled for inter-individual differences in environmental adaptation and psychological status that may modulate perceptual and cognitive responsivity. Seventy-two construction professionals (masons, carpenters, electricians, and concrete workers) satisfying these specifications were recruited (*Appendix Table 3*).

The experimental protocol employed a randomized, counterbalanced, repeated-measures design, with data collection structured across two diurnal sessions: matutinal (09:30–12:00) and postprandial (15:30–17:00). To mitigate olfactory fatigue and constrain experimental duration [52], a balanced incomplete block design (BIBD) was implemented wherein each participant was randomly allocated two of three odorant conditions (*M. piperita, C. sinensis*, and *R. officinalis*). Under each assigned odorant, participants underwent sequential exposure to three acoustic intensities (50, 70, and 85 dBA), yielding six experimental trials per participant. While no individual experienced all nine possible odor–noise permutations, the BIBD ensured complete counterbalancing of treatment combinations across the aggregate sample. Trial presentation order was randomized to minimize sequence effects, and a 5-minute inter-trial washout interval was imposed to attenuate residual sensory carryover. The procedure comprised two sequential phases:

Thermal Equilibration Phase. Following EEG montage and impedance verification, participants completed standardized familiarization (1–2 cognitive task iterations) and baseline demographic assessment. A mandatory 15-minute thermal equilibration period was imposed to establish physiological steady-state prior to stimulus exposure.

Experimental Manipulation Phase. Immediately post-equilibration, olfactory atomization and calibrated construction noise were initiated concomitantly. EEG signals were recorded continuously throughout the 5-minute exposure epoch, followed immediately by cognitive testing and subjective perceptual ratings. This protocol was replicated across six randomized odor–noise permutations with 5-minute inter-trial washout intervals to attenuate sensory carryover (Fig. 2).

2.3. Measurements and data processing

2.3.1. Questionnaire survey

The psychometric instrument comprised two sections: (i) sociodemographic and occupational variables (sex, age, stature, body mass, clothing insulation, occupational classification, and years of service); and (ii) subjective environmental perception indices (*Appendix Fig. A*). Thermal perception assessment included thermal sensation vote (TSV) and thermal comfort vote (TCV). TSV was measured on a seven-point scale (−3: cold, −2: cool, −1: slightly cool, 0: neutral, +1: slightly warm, +2: warm, +3: hot). TCV employed a five-point scale (−2: very uncomfortable, −1: uncomfortable, 0: neutral, +1: comfortable, +2: very comfortable). Auditory sensation vote (ASV, perceived loudness level) used a seven-point scale (−3: very quiet, −2: quiet, −1: slightly quiet, 0: neutral, +1: slightly loud, +2: loud, +3: very loud); auditory comfort vote (ACV, perceived acoustic comfort) used a five-point scale (−2: very uncomfortable, −1: uncomfortable, 0: neutral, +1:

Table 1
Effects of odors on perception and cognitive performance.

Types of odors	Results	References
orange and peppermint	Under the conditions of orange and peppermint (like), subjects experienced an increase in their level of happiness, leading to improved cognitive performance.	[50]
peppermint, lemon and rosemary	Peppermint demonstrates a significant improvement of β waves, whereas rosemary and lemon exhibit a notable increase in θ waves.	[19]
peppermint, lemon and rosemary	Subjective evaluations were significantly different before and after olfactory adaptation, particularly in the rosemary condition. Lemon aroma had the highest preference and led to the highest scores in the memory task.	[32]
rosemary	Rosemary aroma decreased sleepiness and increased alertness in shift-working nurses.	[33]
sweet orange	Short-term indoor exposure to sweet orange essential oil, with the predominant *citrusy* aroma impression, improved subjects' relaxation during mental work breaks.	[35]
peppermint	In terms of subjective mood peppermint increased alertness and ylang-ylang decreased it, but significantly increased calmness.	[38]
peppermint and dry ginger	Peppermint scent is most effective in high-temperature environments, significantly reducing thermal sensation votes (TSV), increasing thermal comfort votes (TCV). Conversely, dry ginger scent is most effective in low-temperature environments, significantly increasing both TSV and TCV,	[51]
peppermint	Peppermint significantly reduced TSV and increased the relative power of α waves	[39]
Lavandula officinalis, Rosa rugosa, and *Mentha canadensis*	*Mentha canadensis* and *Lavandula officinalis* provided an increase in positive affectivity.	[42]

comfortable, +2: very comfortable). Olfactory pleasantness vote (OPV, perceived hedonic valence of olfactory stimuli) was assessed using a seven-point scale (−3: very unpleasant, −2: unpleasant, −1: slightly unpleasant, 0: neutral, +1: slightly pleasant, +2: pleasant, +3: very pleasant). Olfactory comfort vote (OCV, perceived comfort of the olfactory environment) was measured on a five-point scale (−2: very uncomfortable, −1: uncomfortable, 0: neutral, +1: comfortable, +2: very comfortable).

To address potential semantic ambiguities in questionnaire items and accommodate diverse educational backgrounds among construction workers, researchers provided standardized verbal clarifications to ensure conceptual comprehension and maximize the validity of subjective self-reports.

2.3.2. Cognitive test

Neurobehavioral psychometric tests can be used to evaluate the effects on productivity [24,22]. The cognitive ability test adopted in this study has also been used in previous investigations [22,25]. The starting point of the test was labeled "Start: 00″. During the experiment, workers were required to connect numbers in ascending order within a limited time. The accuracy of identifying and connecting the target points was used to analyze workers' cognitive performance. Different numerical sequences were used for each participant.

To eliminate inter-individual differences in cognitive test scores, the speed and accuracy of completing the cognitive test were non-dimensionalized. The nondimensionalization of test performance was conducted based on each participant's own data, as follows:

$$\text{Speed}_{\text{relative}} = \text{Speed}_{\text{i}} / \text{Speed}_{\text{max}} \quad (2)$$

$$\text{Accuracy}_{\text{relative}} = \text{Accuracy}_{\text{i}} / \text{Accuracy}_{\text{max}} \quad (3)$$

where $Speed_i$ and $Accuracy_i$ represent the absolute completion speed and accuracy of the cognitive test for each participant under experimental condition i, respectively. $Speed_{max}$ and $Accuracy_{max}$ denote the maximum speed and accuracy achieved by the participant across all experimental conditions.

2.3.3. EEG monitoring

Neurophysiological recordings were acquired using a 14-channel Emotiv EPOC X wireless EEG system (EMOTIV Inc., USA), incorporating common mode sense (CMS) and driven right leg (DRL) reference electrodes to ensure signal fidelity. Scalp electrodes were positioned according to the International 10–20 System at AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4, providing comprehensive coverage of frontopolar, frontal, frontocentral, temporal, parietal, and occipital cortical territories. Data were sampled continuously at 128 Hz. Prior to montage, sensors were saturated with 5% saline solution to optimize scalp-electrode impedance. The headset was fitted by trained researchers, with stable electrode-scalp contact verified and maintained throughout the experimental duration. Neurophysiological data were recorded and archived in real-time using Emotiv PRO software [53]. (Fig. 3).

Continuous EEG recordings were preprocessed using EEGLAB (version 14.0.0) running within the MATLAB environment. The analytical pipeline encompassed initial artifact screening via visual inspection, followed by bandpass filtering (1–30 Hz, FIR) to isolate δ, θ, α, and β oscillations while attenuating ultra-low frequency drift and high-frequency muscular contaminants. Independent Component Analysis (ICA) was subsequently applied to isolate and attenuate physiologic

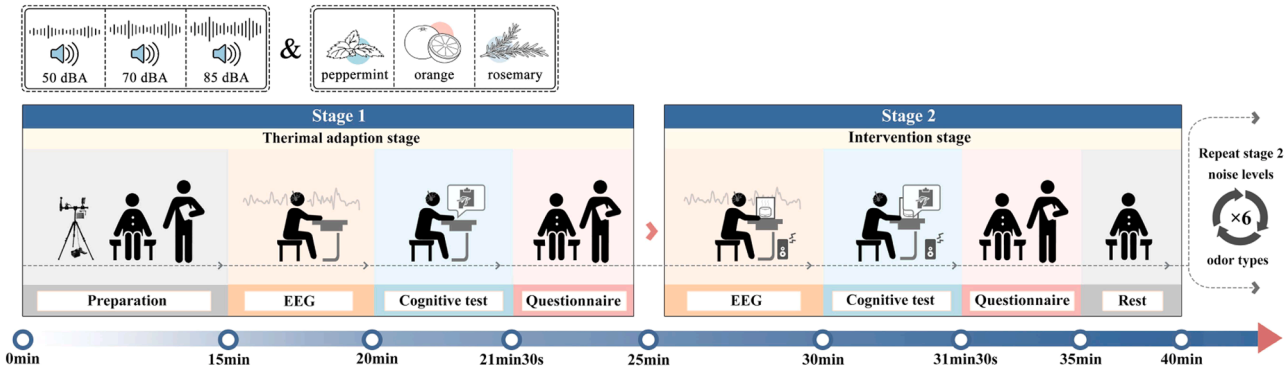


Fig. 2. Experimental procedure.

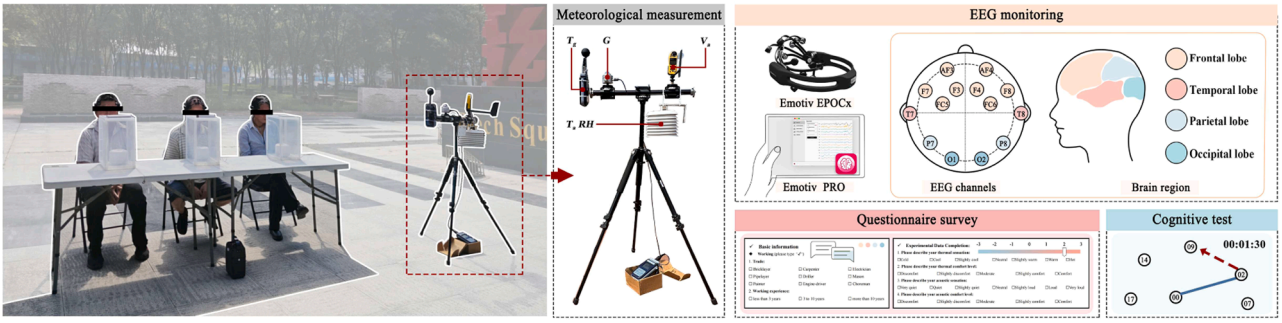


Fig. 3. Schematic of the trial setup.

artifacts, particularly electrooculogram (EOG) and myogenic activity, identified via component topography and temporal dynamics. Residual cardiac artifacts were automatically detected and rejected using the Automated Artifact Detection toolbox (ADJUST) for EEGLAB.

Neural oscillatory activity was operationalized via power spectral density (PSD) analysis. PSD was derived from preprocessed epochs using the Fast Fourier Transform (FFT) algorithm, quantifying signal power distribution ($\mu V^2/Hz$) across the frequency domain to characterize spectral band dynamics.

$$F[k] = \sum_{n=0}^{N=1} x[n] e^{\frac{-j2\pi kn}{N}} K = 0, \ldots, N-1 \quad (4)$$

$$P(k) = \lim_{n\to\infty} \left(\frac{|F(k)|}{N} \right)^2 \quad (5)$$

where F is the Fourier transform, x is the signal, and P is the power spectral density.

Normalized spectral power densities (relative power indices) were computed across canonical frequency bands (δ, θ, α, β) for all 14 electrode channels to minimize inter-individual amplitude variability inherent in absolute power metrics. Group-level scalp topographies were subsequently generated to visualize spatial distributions of band-specific oscillatory activity. The normalisation procedure was implemented as follows:

$$P_{nor} = \frac{P_i}{\sum_{i\in(\delta,\theta,\alpha,\beta)} P_i} \quad (6)$$

where P_i represents the absolute power of each frequency band, and the normalized power spectrum is defined as the ratio of the absolute power of each band to the sum of the absolute powers of the four bands (δ, θ, α, and β).

2.3.4. Thermal stress category

Outdoor thermal stress was quantified using the PET, a thermophysiological index derived from the Munich Energy Balance Model for Individuals (MEMI) that integrates all major heat exchange processes to express complex outdoor thermal conditions in terms of equivalent indoor reference temperatures [54,55]. Meteorological parameters (T_a, RH, V_a, and T_{mrt}) together with participant-specific anthropometric and physiological attributes (stature, body mass, age, sex, clothing insulation, and metabolic rate) were input into the RayMan model to derive PET values for each experimental condition.

Subsequent analyses were performed using Microsoft Excel and Origin 2018. Linear regression analysis was conducted to establish the relationship between PET and aggregated mean thermal sensation votes (MTSV) across 1 °C PET intervals. Based on established thermal perception thresholds [56], experimental conditions were stratified into three distinct thermal stress categories: 17.36 °$C \leq$ PET $<$ 32.43 °C (Neutral, 0.5 $\leq$ TSV $<$ 0.5), 32.43 °$C \leq$ PET $<$ 47.51 °C (Slightly warm, 0.5 $\leq$ TSV $<$ 1.5), and 47.51 °$C \leq$ PET $<$ 62.59 °C (Warm, 1.5 $\leq$ TSV $<$2.5) (Fig. 4).

2.4. Statistical analysis

Statistical analyses were performed using IBM SPSS Statistics 26.0 and Microsoft Excel 2017. Descriptive distributions were visualized using percentage stacked bar charts. Linear mixed models (LMMs) were fitted to assess main and interaction effects of thermal stress, acoustic intensity, and odorant type on subjective perception and EEG spectral power. Participant identification was specified as a random intercept to account for within-subject correlations [57]. Covariates (body mass index, age, and professional tenure) were initially incorporated but excluded from final models due to non-significance (all $p > 0.50$), ensuring parsimony. Significant interactions were decomposed via simple effects analyses. Cognitive performance was analyzed using Kruskal–Wallis tests (given violations of normality and homoscedasticity) and illustrated using box-and-whisker plots. The relative efficacy of multimodal interventions was evaluated using the entropy-weighted TOPSIS (EW-TOPSIS) method. This multi-criteria framework calculates Euclidean distances from positive- and negative-ideal solutions to derive relative closeness coefficients (C_i; range 0–1), establishing Pareto-optimal rankings (*Appendix Method*).

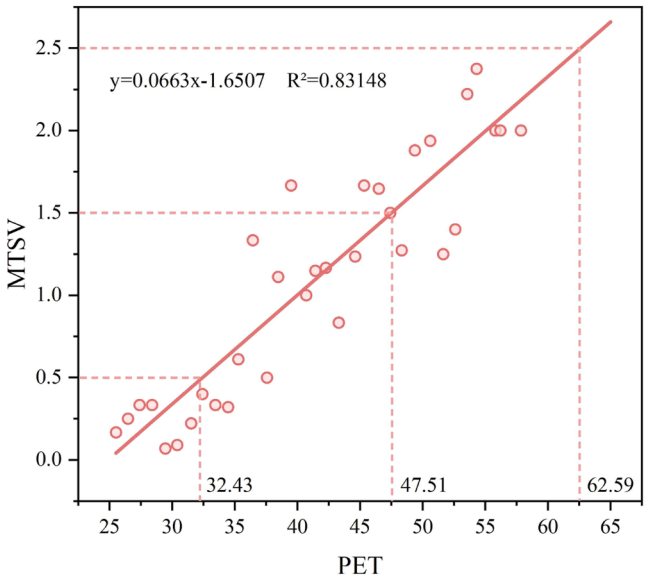


Fig. 4. Correlation between PET and MTSV.

3. Results

3.1. Subjective perception

Elevated acoustic stress systematically shifted TSV toward warmer perceptions across all thermal stress strata (Fig. 5a). Specifically, under "neutral" conditions, the prevalence of TSV = 0 declined significantly from 87.33% (50 dBA) to 72.55% (85 dBA), with no instances of TSV = 3 recorded at any acoustic level. Conversely, under "warm" conditions, the incidence of TSV = 3 increased monotonically with noise intensity, rising from 13.33% at 50 dBA to 29.63% and 40.74% at 70 dBA and 85 dBA, respectively. Concomitantly, TCV exhibited systematic deterioration with escalating noise exposure (Fig. 5b). Within the "neutral" regime, the proportion of TCV = 2 decreased from 23.40% (50 dBA) to 11.76% (85 dBA). Under "slightly warn" conditions, discomfort reports (TCV = –2) increased from 2.44% to 6.06%, evincing a synergistic amplification of thermal discomfort under combined thermal and acoustic stressors.

Thermal stress amplification exacerbated auditory sensitivity (Fig. 5c). Specifically, under high-intensity acoustic exposure (85 dBA), the prevalence of ASV = 2 increased from 11.76% under "neutral" conditions to 33.33% under "warm" conditions. Concomitantly, elevated thermal stress intensified auditory discomfort (Fig. 5d). Relative to thermoneutral baselines, high-noise conditions (85 dBA) yielded incremental increases in uncomfortable auditory sensations (ACV = –1) of 15.18% and 22.26% under "slightly warm" and "warm" conditions, respectively.

The OPV exhibited acoustic-invariant distribution patterns (Fig. 5e). Under "slightly warm" conditions, the proportions reporting "OPV = 1″ remained remarkably stable across acoustic intensities (26.82%, 27.27%, and 28.78% at 50, 70, and 85 dBA, respectively). By contrast, acoustic exposure attenuated OCV selectively under "neutral" conditions, with proportion reporting "OCV = 1" declining from 46.81% (50 dBA) to 39.22% (85 dBA); this cross-modal modulatory effect was, however, suppressed under elevated thermal stress (Fig. 5f).

Within the same thermal stress stratum, olfactory stimuli induced differential modulation of thermal sensation (Fig. 6a). Under "neutral" and "slightly warm" conditions, "TSV = 0" predominated across all odorant exposures; however, *C. sinensis* elicited the lowest prevalence (67.92% and 40.30%), followed by *R. officinalis* (85.71% and 49.30%) and *M. piperita* (87.04% and 44.00%). Irrespective of odorant type, thermal sensation escalated with increasing thermal stress. Specifically, hot sensation (TSV = 3) under *M. piperita* increased from 4.00% to 20.00% when transitioning from slightly warm to warm conditions, whereas *R. officinalis* exhibited a more pronounced escalation (8.45% to 43.33%). Concordantly, thermal comfort deterioration (TCV = –2) under *R. officinalis* demonstrated the most marked increment (6.97% to 53.33%) across the three odorants during this thermal transition (Fig. 6b), indicating that *R. officinalis* amplified thermal discomfort under high thermal stress.

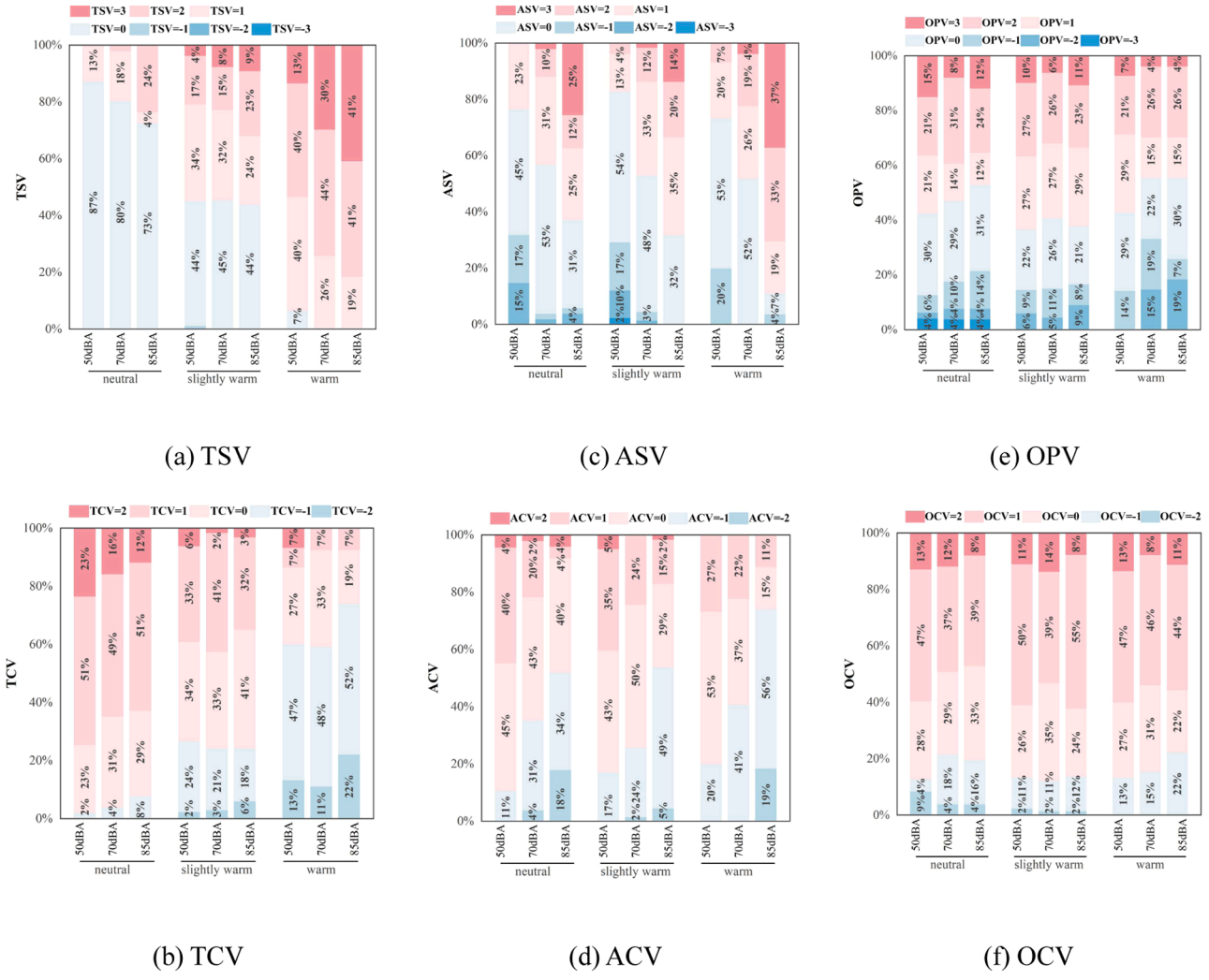


Fig. 5. Distribution of TSV (a), TCV (b), ASV (c), ACV (d), OPV (e), and OCV (f) among noise levels.

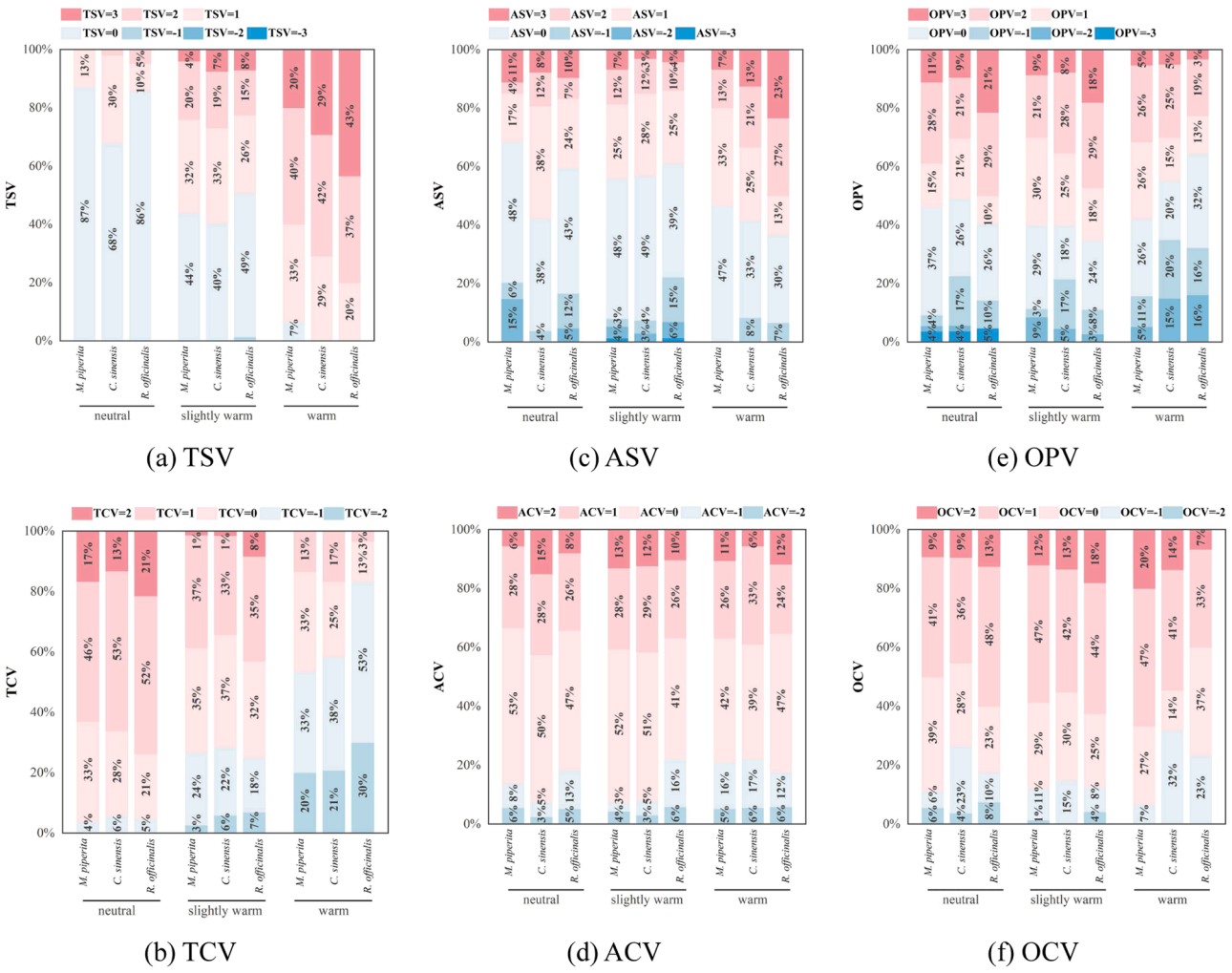


Fig. 6. Distribution of TSV (a), TCV (b), ASV (c), ACV (d), OPV (e), and OCV (f) among odor types.

The ASVs exhibited odor-dependent thermal amplification (Fig. 6c). Transitioning from "neutral" to "slightly warm" conditions increased loudness perception (ASV = 2) by 9.63%, 9.29%, and 19.52% for *M. piperita, C. sinensis*, and *R. officinalis*, respectively. Conversely, the ACVs distributions remained odor-invariant (Fig. 6d). Under "neutral" and "slightly warm" conditions, *C. sinensis* was associated with the lowest prevalence of severe auditory discomfort (ACV = –2), which increased marginally from 2.50% to 3.08% with escalating thermal stress.

R. officinalis consistently evoked the highest "very pleasant" ratings (OPV = 3) under "neutral" (21.42%) and "slightly warm" (18.05%) conditions, significantly exceeding *M. piperita* (11.11% and 8.57%) (Fig. 6e). However, under "warm" conditions, *M. piperita* exhibited the highest "very pleasant" ratings (5.26%) and concurrently the lowest "unpleasant" ratings (5.26%). This thermal-dependent reversal was mirrored in OCV ratings. Under "slightly warm" conditions, *R. officinalis* elicited superior comfort (18.06%) relative to *M. piperita* (12.00%) and *C. sinensis* (13.43%). Yet upon transition to "warm" conditions, *R. officinalis* comfort precipitously declined to 6.67%, whereas *M. piperita* comfort increased to 20.00% (Fig. 6f). These findings demonstrate that while *R. officinalis* exhibits superior hedonic properties under "neutral" to "warm" conditions, its efficacy deteriorates significantly under thermal stress, whereas *M. piperita* maintains or even enhances perceptual comfort under extreme thermal loads.

Linear mixed models demonstrated significant main effects of thermal stress on TSV, TCV, and ASV ($p < 0.01$), and of noise level on ASV, ACV, and TSV ($p < 0.05$). Odorant type did not significantly modulate any subjective perceptual index ($p > 0.05$). Notably, significant thermal stress × odor type interactions emerged for TCV and OPV ($p < 0.05$), necessitating simple-effects decomposition (*Appendix Table 4*). Simple-effects analysis revealed that progressive thermal stress significantly attenuated TCV across all odorant conditions ($p < 0.05$). Specifically, under "warm" condition, TCV was significantly lower for *R. officinalis* relative to *C. sinensis* ($p < 0.05$; Fig. 7a). *R. officinalis* elicited significantly higher OPV ratings under "neutral" and "slightly warm" conditions ($p < 0.001$ and $p < 0.01$, respectively), evincing a marked deterioration in perceived pleasantness with escalating thermal load. By contrast, OPV for *M. piperita* and *C. sinensis* remained stable across thermal stress levels ($p > 0.05$; Fig. 7b).

3.2. Cognitive test

Thermal stress significantly modulated the dose–response relationship between acoustic exposure and psychomotor velocity. Under "neutral" conditions, reaction time distributions exhibited rightward skewness (Mdn = 0.87) at 50 dBA relative to 70 dBA (Mdn = 0.80) and 85 dBA (Mdn = 0.78), indicating that minimal acoustic load optimizes processing speed in thermally neutral environments. Conversely, under "slightly warm" and "warm" conditions, distributions at 70 and 85 dBA demonstrated pronounced rightward shifts, suggestive of noise-facilitated task execution under thermal duress. Notably, the 50 dBA condition exhibited thermal invariance in speed distributions across

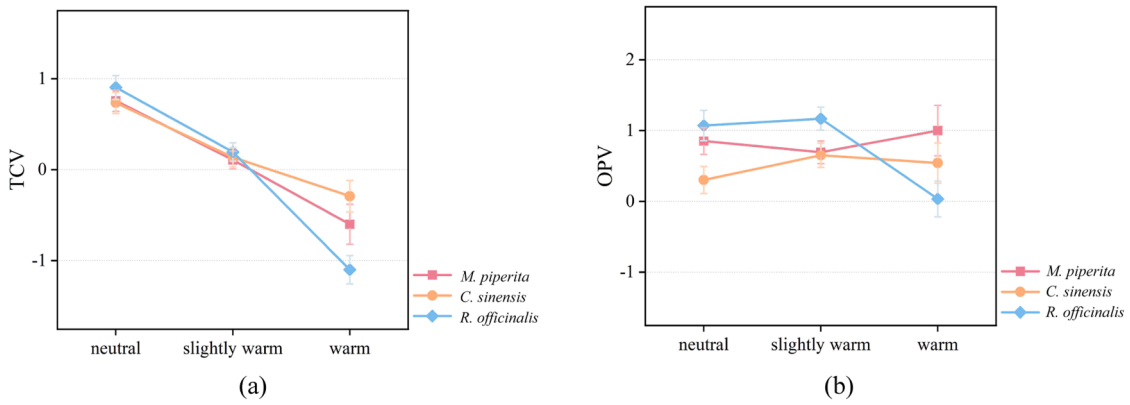


Fig. 7. Impacts of interactions on TCV (a) and OPV (b).

stress levels (Fig. 8a).

Task accuracy, however, manifested an inverted-U relationship across the acoustic arousal continuum, independent of thermal regime. Distributions at 70 dBA consistently concentrated at higher performance values (right-skewed), whereas both 50 and 85 dBA yielded dispersed, left-skewed distributions toward lower accuracy values. This acoustic modulation profile is consistent with the Yerkes–Dodson law, wherein moderate noise facilitates optimal arousal for sustained attention tasks (Fig. 8b).

Thermal stress significantly modulated the differential effects of odorant exposure on psychomotor velocity. With escalating thermal load, velocity distributions under *M. piperita* and *C. sinensis* exhibited pronounced rightward shifts toward higher performance values, whereas *R. officinalis* demonstrated thermal stability in velocity distributions. Specifically, under "slightly warm" and "warm" conditions, *M. piperita* elicited the most favorable velocity profiles (Mdn = 0.87 and 0.90, respectively), while *R. officinalis* showed left-skewed distributions concentrated at lower velocities (Mdn = 0.82 and 0.86) (Fig. 9a). These findings indicate that under thermal duress, *M. piperita* facilitates superior processing speed relative to *R. officinalis*.

Conversely, accuracy distributions exhibited an odor-specific thermal resilience pattern. Across all thermal stress strata, *R. officinalis* consistently yielded right-skewed accuracy distributions concentrated at higher values. By contrast, *M. piperita* produced the most dispersed, left-skewed accuracy patterns with elevated proportions of low-performance outliers. *C. sinensis* demonstrated context-dependent variability, exhibiting dispersed distributions specifically under "slightly warm" conditions (Fig. 9b). Collectively, these results demonstrate that while *M. piperita* optimizes psychomotor velocity under thermal stress, *R. officinalis* confers superior accuracy maintenance, evincing a speed–accuracy trade-off modulated by both odorant type and concentration.

Kruskal–Wallis tests revealed significant main effects of thermal stress on psychomotor velocity ($p < 0.01$) and of acoustic exposure on task accuracy ($p < 0.05$). Odorant type did not significantly modulate either performance dimension ($p > 0.05$). Post hoc pairwise comparisons indicated that velocity under "warm" and "slightly warm" conditions significantly exceeded "neutral" condition ($p < 0.01$). For acoustic conditions, accuracy at 70 dBA significantly surpassed both suboptimal (50 dBA) and supraoptimal (85 dBA) noise levels ($p < 0.05$) (Table 2).

Collectively, these findings indicate that simple psychomotor tasks exhibit facilitated processing speed under escalating thermal stress, consistent with arousal-mediated performance enhancement. Furthermore, moderate acoustic stimulation (70 dBA) optimizes accuracy performance relative to both hypo- and hyper-arousing noise levels, corroborating the inverted-U relationship between environmental arousal and cognitive efficiency.

3.3. Electroencephalogram

Group-averaged topographic mappings of spectrally decomposed and normalized data (δ, θ, α, β) delineate cortical oscillatory patterns evoked by multimodal stressors, constituting neurophysiological indices of arousal and affect. Chromatic coding indicates relative power density (red spectrum: enhanced; blue spectrum: attenuated oscillatory activity).

Progressive thermal loading evoked frequency band-specific modulations indicative of central nervous system depression. Escalating thermal stress significantly amplified δ-band spectral power across frontoparietal, temporal, and occipital cortices, concomitant with α- and β-band power attenuation. This frequency-shift pattern, which is characterized by increased slow-wave dominance and reduced fast-wave activity, constitutes a neurophysiological signature of drowsiness and diminished vigilance (Fig. 10a). Increasing noise exposure induced frontocortical hypoactivation, manifesting as reduced α- and β-band spectral power within frontal electrode sites. This topographic pattern

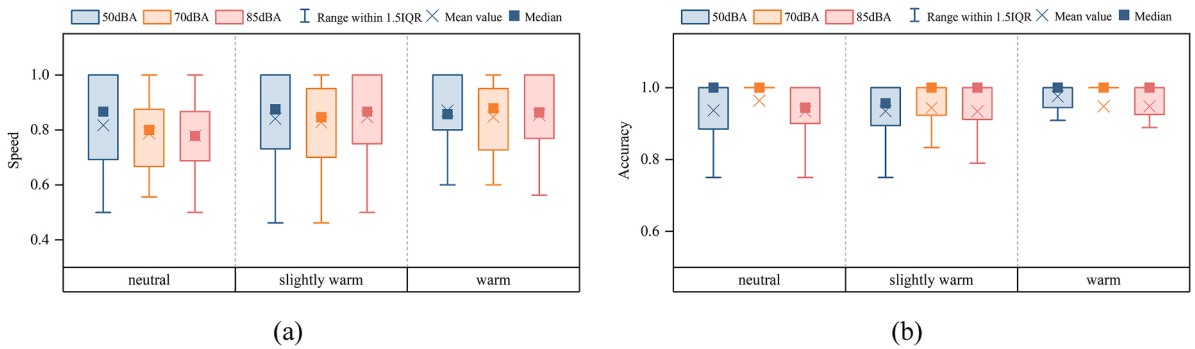


Fig. 8. Speed and Accuracy in different thermal stress and noise levels: Speed (a); Accuracy (b).

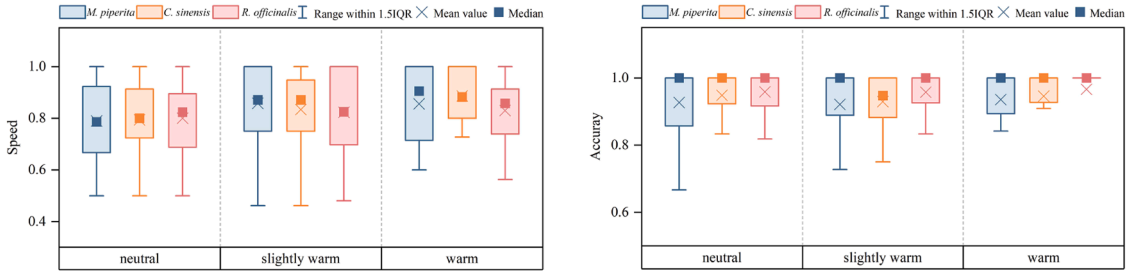


Fig. 9. Speed and Accuracy in different thermal stress and olfactory types: Speed (a); Accuracy (b).

Table 2
Effects of environmental factors on Speed and Accuracy.

Factors	Dependent variables	Chi-squares	df	p	Pairwise Comparisons	p
Thermal stress	Speed	11.138	2	0.004	warm>neutral	0.004
					slightly warm> neutral	0.004
	Accuracy	3.612	2	0.164	NaN	
Noise level	Speed	1.838	2	0.399	NaN	
	Accuracy	6.136	2	0.047	70dBA>50dBA	0.030
					70dBA>85dBA	0.029
Odor type	Speed	0.658	2	0.720	NaN	
	Accuracy	5.733	2	0.057	NaN	

suggests that elevated acoustic stress compromises both parasympathetic-mediated relaxation and attentional alertness (Fig. 10b). Distinct odorants elicited regionally specific neurophysiological signatures. *M. piperita* engendered widespread β-band enhancement across frontocortical, temporal, and occipital regions, suggesting robust cortical arousal facilitation. *C. sinensis* was associated with isolated frontal θ-band elevation concurrent with β-band suppression across multiple lobes, indicating prefrontal affective processing with concomitant vigilance decrement. Conversely, *R. officinalis* produced dual topographic patterns: α-band augmentation in frontal regions and β-band enhancement in parietal cortex, suggesting a dissociation between relaxation and arousal across distinct cortical territories (Fig. 10c).

Linear mixed models were fitted to assess environmental effects on region-specific oscillatory dynamics across δ, θ, α, and β bands (*Appendix Table 5*). Thermal stress exerted robust main effects on δ- and β-band power across frontocortical, temporal, parietal, and occipital regions ($p < 0.001$), and significantly modulated α-band activity in temporal, parietal, and occipital cortices ($p < 0.05$). Acoustic exposure selectively influenced frontocortical α-band power ($p < 0.05$) without significant effects in other bands or regions. Odorant type significantly modulated θ-band activity in frontal and occipital cortices ($p < 0.05$).

Furthermore, significant thermal stress × odorant interactions were observed for β-band power across frontal, temporal, and occipital regions, and for δ-band activity in the temporal lobe ($p < 0.05$), necessitating simple-effects decomposition. Simple-effects analysis indicated that under "neutral" conditions, *M. piperita* induced significantly elevated β-band power across frontal, temporal, and occipital cortices relative to both *C. sinensis* and *R. officinalis* ($p < 0.05$), indicative of superior cortical arousal facilitation. However, this arousal-enhancing effect exhibited thermal fragility; escalating thermal stress from "neutral" to "slightly warm" significantly attenuated β-band power across these regions, concomitant with augmented occipital δ-band activity ($p < 0.05$). Conversely, *C. sinensis* exposure resulted in significant β-band attenuation in temporal and occipital regions when transitioning from "neutral" to "warm" conditions, accompanied by further temporal δ-band augmentation under escalating thermal stress (from "slightly warm" to "warm"). Notably, *R. officinalis* exhibited a distinct neurophysiological resilience profile: significant β-band attenuation was restricted exclusively to the occipital cortex under "warm" relative to "neutral" conditions ($p < 0.05$). This restricted topographic vulnerability suggests *R. officinalis* confers greater neurophysiological stability and resistance to thermal stress–induced cortical dysregulation relative to alternative odorants (Fig. 11).

3.4. Comprehensive score

Given the context-dependent modulation of noise–odor interactions across thermal stress strata, entropy-weighted TOPSIS multi-criteria decision analysis was employed to derive Pareto-optimal rankings of multimodal intervention efficacy. The evaluation matrix comprised psychophysical indices (TSV, TCV, ASV, ACV, OPV, OCV), cognitive performance metrics (accuracy), and neurophysiological markers (relative spectral power of δ, θ, α, and β oscillations). All indicators underwent benefit-type orientation transformation and min–max normalization prior to objective weighting via information entropy theory, followed by computation of relative closeness indices (C_i) to ideal solutions.

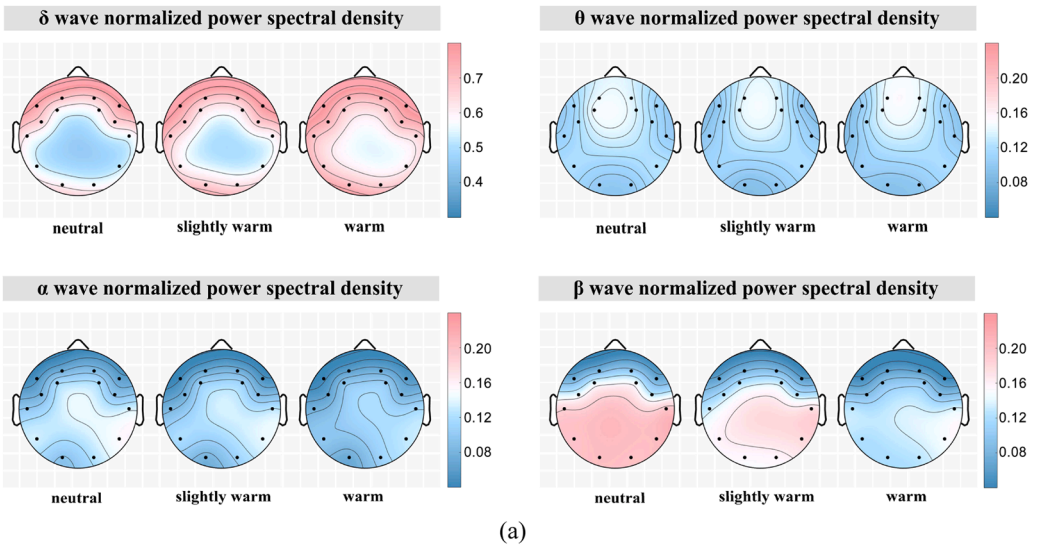
Entropy analysis revealed temporal β-band spectral power as the principal discriminating criterion ($\omega = 0.0986$), followed by thermal comfort votes ($\omega = 0.0964$) and task accuracy ($\omega = 0.0882$). Aggregated rankings demonstrated a general deterioration of high-noise combinations with escalating thermal load, with odor-specific efficacy varying markedly across thermal regimes.

Specifically, under "neutral" conditions, the low-noise/*R. officinalis* combination achieved optimal integrated efficacy ($C_i = 0.738$), whereas the high-noise/*C. sinensis* combination exhibited the poorest performance ($C_i = 0.421$). Under "slightly warm" conditions, the medium-noise/*R. officinalis* combination ranked highest ($C_i = 0.595$), while the high-noise/*M. piperita* combination demonstrated minimal efficacy ($C_i = 0.367$). Notably, under "warm" conditions, the ranking hierarchy inverted: the low-noise/*M. piperita* combination emerged as optimal ($C_i = 0.446$), whereas the high-noise/*M. piperita* combination remained the least efficacious ($C_i = 0.232$) (Fig. 12).

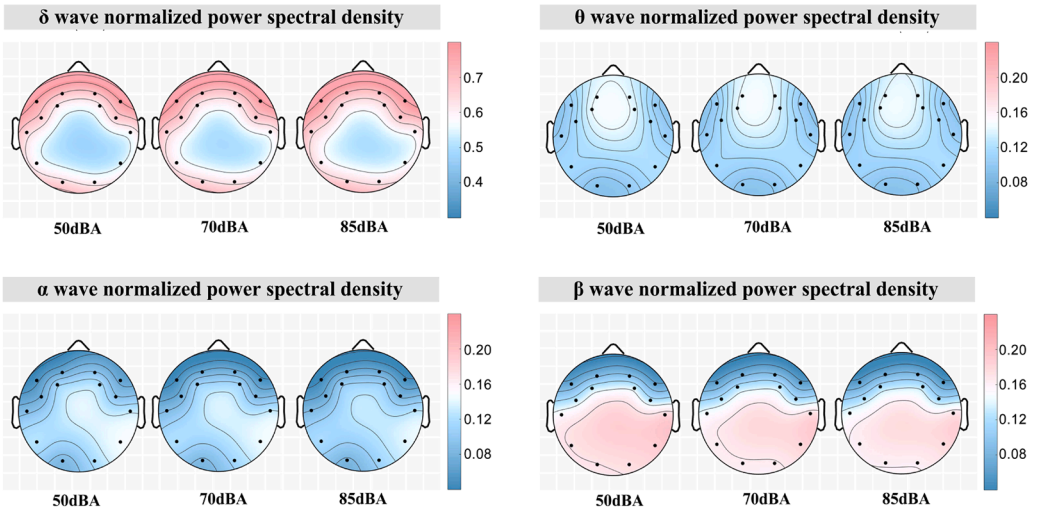
4. Discussion

4.1. Subjective perception

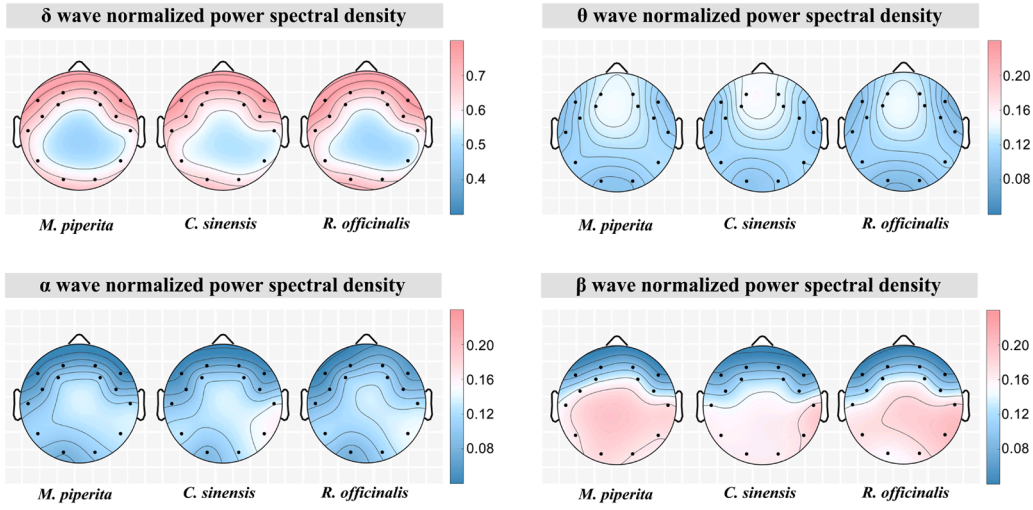
Our findings demonstrate significant cross-modal interaction effects: acoustic exposure amplified TSV ($p < 0.05$), while thermal stress intensified ASV ($p < 0.01$), corroborating previous psychophysical evidence [58]. Notably, we observed no significant modulation of ACV by thermal stress, nor of TCV by noise level—findings that diverge from prior investigations [47,59]. We posit that this discrepancy reflects occupational adaptation: our cohort comprised construction workers with chronic environmental exposure, exhibiting elevated tolerance



(a)



(b)



(c)

Fig. 10. Topographical distributions and developments of δ, θ, α, and β wave normalized power spectral density: thermal stress (a), noise level (b), and odor type (c).

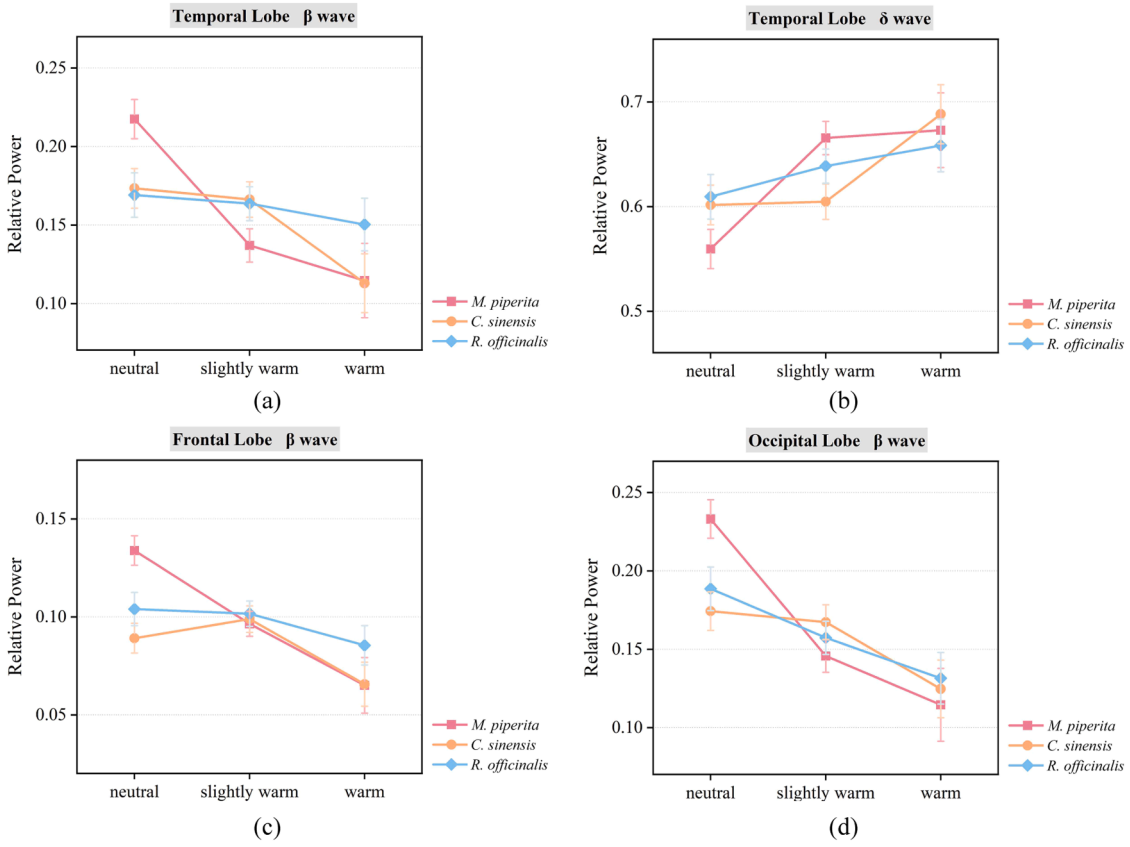


Fig. 11. Impacts of interactions on frontal lobe β wave (a), temporal lobe δ wave (b), temporal lobe β wave (c), and Occipital lobe δ wave (d).

thresholds and allostatic load resistance compared to the university student populations employed in previous studies [60].

Under "warm" conditions, *C. sinensis* elicited superior OPV and TCV relative to *R. officinalis* ($p < 0.05$). This divergence likely stems from differential neuropharmacological mechanisms: limonene, the predominant monoterpene in orange essential oil, modulates anxiolytic pathways via GABAergic, serotonergic, and dopaminergic neurotransmission [61], thereby enhancing hedonic tone and perceptual comfort [50]. Conversely, *R. officinalis* primarily engages cholinergic and glutamatergic circuits associated with cognitive enhancement rather than affective regulation [62]. Consequently, under thermal duress, the affective rebound of discomfort may suppress olfactory hedonic responses to *R. officinalis*, whereas *C. sinensis* maintains its anxiolytic efficacy [42].

4.2. Cognitive test

Thermal stress significantly facilitated psychomotor velocity ($p < 0.01$), corroborating previous evidence that escalating thermal load induces compensatory behavioral acceleration to minimize exposure duration and mitigate physiological strain [24,63]. This expedited responding likely reflects a motivated behavioral adaptation strategy wherein participants prioritize task completion to alleviate thermal discomfort [64].

Conversely, acoustic exposure exhibited an inverted-U relationship with cognitive precision ($p < 0.05$), wherein accuracy at 70 dBA significantly exceeded both suboptimal (50 dBA) and supraoptimal (85 dBA) conditions. This finding diverges from Chong et al. [65], who reported linear performance decrements across 45–85 dBA. We attribute this discrepancy to ecological validity and occupational adaptation: construction workers chronically exposed to dynamic, high-amplitude outdoor soundscapes likely exhibit attenuated auditory sensitivity and enhanced signal-to-noise filtering capacity relative to participants tested in stable indoor environments.

Notably, odorant type failed to significantly modulate either velocity or accuracy metrics. This null result contrasts with Shi et al. [19], likely attributable to differential attentional resource allocation under composite environmental stress. In the present ecological paradigm, limited cognitive capacity may be monopolized by dominant thermal and acoustic stressors, effectively masking subthreshold olfactory modulation of executive function. Alternatively, this absence of effect may reflect concentration-specific dose–response relationships; extant literature indicates that essential oil efficacy exhibits narrow therapeutic windows, with cognitive benefits manifesting only within precise concentration ranges that may not have been achieved under the present field conditions [19].

4.3. Electroencephalogram

Progressive thermal loading elicited a characteristic neurophysiological signature of hypoarousal: significant augmentation of δ-band power concomitant with attenuation of α- and β-band oscillations across cortical territories. This spectral shift pattern is consistent with established neural markers of diminished vigilance and executive control [66,67]. Regarding acoustic stress, elevated noise exposure (85 dBA) significantly suppressed frontocortical α-band power relative to low acoustic levels (50 dBA) ($p < 0.05$), corroborating previous indoor investigations [65]. Given the association between α oscillations and parasympathetic relaxation states, this attenuation likely reflects heightened psychological stress and cortical hyperexcitability. The frontal cortex demonstrates particular vulnerability to transient functional dysregulation under acute high-intensity acoustic exposure, with chronic exposure potentially compromising neural integrity [68].

Under "neutral" conditions, odorant-specific neurophysiological signatures emerged. *M. piperita* elicited significantly elevated β-band

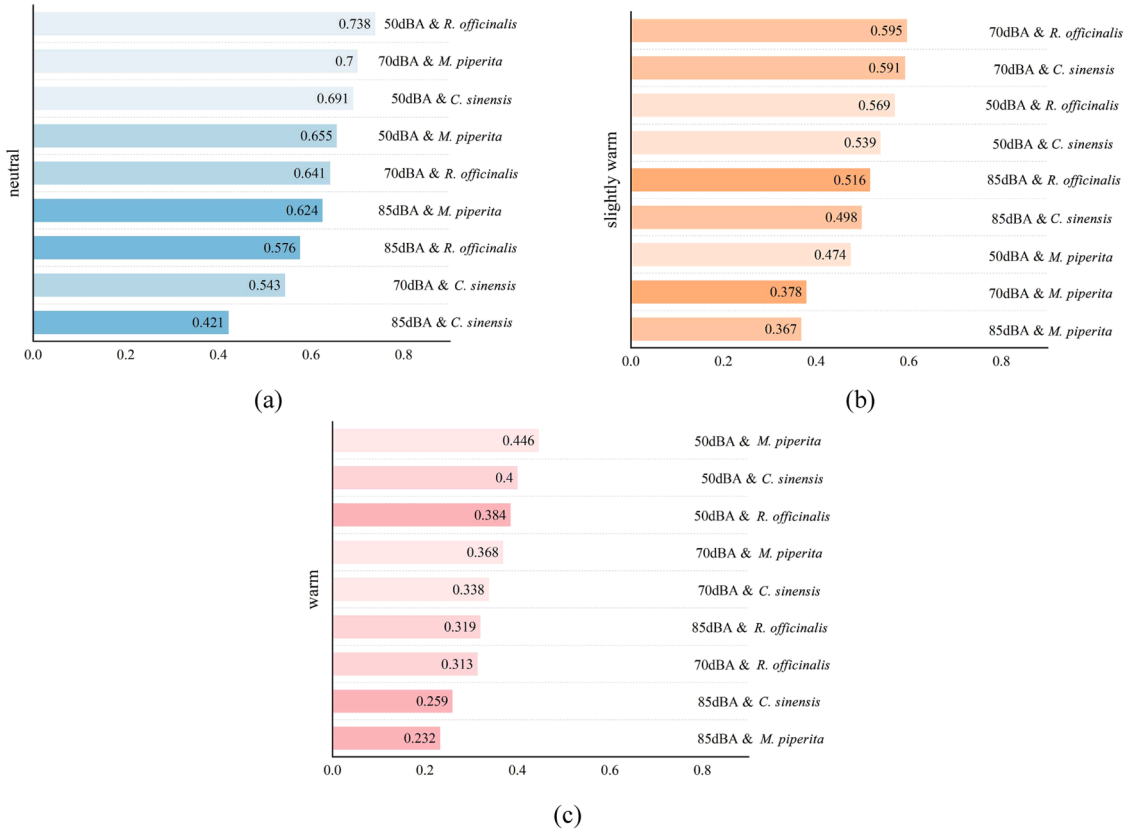


Fig. 12. Ranking of the comprehensive benefits of different environmental combinations under three thermal stress levels: neutral (a), slightly warm (b), and warm (c) conditions.

power across frontocortical, temporal, and parietal regions relative to alternative odorants ($p < 0.05$), whereas *C. sinensis* induced enhanced θ-band activity in frontal and occipital cortices compared to *R. officinalis* ($p < 0.05$). *R. officinalis*, conversely, exhibited selectively elevated parietal β-band power relative to *C. sinensis* ($p < 0.05$). These topographic patterns align with extant literature demonstrating that *M. piperita* and *R. officinalis* facilitate cortical arousal and sustained attention, whereas *C. sinensis* promotes anxiolytic relaxation [19,35,38,69].

However, this odorant-specific neurophysiological differentiation exhibited thermal fragility. Transitioning from "neutral" to "warm" conditions precipitated widespread β-band attenuation and δ-band augmentation across most odor conditions, indicative of thermally-induced hypoarousal. Critically, *R. officinalis* demonstrated relative neurophysiological resilience, manifesting minimal β-band decrements across cortical regions. This stability may be attributed to the neuroprotective actions of rosmarinic acid—a bioactive constituent of *R. officinalis*—which mitigates cellular stress-induced neuronal damage [70,71], exerts antioxidant effects in hippocampal and prefrontal structures [72], and attenuates cerebral reactive oxygen species (ROS) generation [73]. Such mechanisms may confer partial protection against thermal stress–induced suppression of high-frequency cortical oscillatory activity.

4.4. Applications

Implementation of a dynamic, multi-criteria occupational health surveillance system is imperative, integrating real-time monitoring of PET and acoustic exposure to trigger context-specific interventions. The following stratified management protocols are recommended based on thermal stress gradients:

Under "neutral" conditions (17.36≤PET<32.43 °C), it supports optimal neurophysiological arousal and is conducive to cognitively demanding operations requiring sustained attention. Under low acoustic load (50 dBA), *R. officinalis* delivery via personal diffusion systems can enhance concentration. However, elevated noise levels (85 dBA) significantly compromise accuracy; therefore, engineering noise controls and low-decibel construction methodologies should be prioritized. When high-intensity noise is unavoidable, workers must utilize certified personal protective equipment (PPE), supplemented by *M. piperita* aerosol delivery via HVAC or misting systems to mitigate acoustic-induced vigilance decrements.

Under "slightly warm" conditions (32.43≤PET<47.51 °C), neurophysiological markers indicate elevated fatigue and attenuated alertness, necessitating enhanced microclimate cooling, frequent short-duration breaks, and adequate hydration protocols. Notably, *M. piperita* exhibits minimal integrated efficacy under this thermal load and should be deployed cautiously. Conversely, personalized *R. officinalis* delivery may sustain attentional resources through α-band modulation. For recreational rest zones, *C. sinensis* is recommended to optimize hedonic tone and psychophysiological recovery.

Under "warm" conditions (47.51≤PET<62.59 °C), mandatory heat alerts should be issued, high-intensity physical labor suspended, and concurrent high-decibel operations minimized. Under low acoustic load, integrated *M. piperita* delivery with convective cooling and misting systems yields optimal integrated efficacy scores. However, under elevated acoustic stress (85 dBA), *M. piperita* efficacy precipitously declines; its use under such dual-stress conditions requires careful risk-benefit assessment.

4.5. Limitation and further directions

Our study exists some limitations: 1) this research elucidated the multidimensional regulatory effects of phytogenic olfactory interventions on construction workers' neurobehavioral performance

across subjective, cognitive, and neurophysiological domains in the continental cold climatic region. Future investigations should extend this framework to hot–humid zones to ascertain whether olfactory efficacy exhibits climatic dependency. 2) The dose–response characteristics of high-arousal odorants remain uncharacterized; systematic manipulation of odor concentration is required to establish optimal therapeutic windows. 3) Potential sex-based disparities in psychophysiological responsivity should be examined to inform gender-sensitive intervention strategies, despite existing literature suggesting minimal differences [74]. 4) Integration of multimodal physiological indices (sudomotor activity, cardiac autonomic function, and electrodermal responses) would enable comprehensive characterization of autonomic and thermoregulatory correlates of olfactory stimulation in occupational outdoor settings. Such multi-system assessments would significantly advance our understanding of peripheral–cephalic coupling in environmental health modulation. 5) While intentionally constrained to a trimodal (thermal–acoustic–olfactory) paradigm to ensure methodological rigor and evaluate feasible countermeasures for thermal and acoustic stress, we acknowledge that authentic construction environments encompass multifactorial exposures (e.g., particulate matter, luminous pollution). Future investigations should broaden this framework by integrating additional ambient stressors to enhance ecological validity and reflect complex occupational settings.

5. Conclusions

This field experiment was conducted at an urban plaza and pavilion in Xi'an, China, spanning three thermophysiological stress levels. Seventy-two construction workers underwent randomized exposure to factorial combinations of three acoustic intensities (50, 70, and 85 dBA) and three phytogenic odorants (*M. piperita, C. sinensis*, and *R. officinalis*). Neurobehavioral responses were assessed via micrometeorological monitoring, psychometric evaluation, cognitive psychomotor testing, and quantitative EEG. The principal findings are as follows:

(1) Multimodal thermal–acoustic stress elicited convergent perceptual–neurophysiological signatures. Thermal stress significantly amplified TSV and attenuated TCV ($p < 0.001$), whereas acoustic stress intensified ASV and degraded ACV ($p < 0.001$), with significant cross-modal facilitation of discomfort ($p < 0.05$). At the neurophysiological level, thermal stress induced widespread δ-band augmentation and β-band attenuation across frontocortical, temporal, parietal, and occipital regions, concomitant with α-band suppression in temporal, parietal, and occipital cortices (all $p < 0.05$). Acoustic stress selectively attenuated frontocortical α-band power ($p < 0.05$).

(2) While odorant type failed to significantly influence subjective perception or cognitive performance ($p > 0.05$), distinct neurophysiological signatures were elicited ($p < 0.05$). Specifically, *C. sinensis* significantly elevated θ-band activity in frontal and occipital regions relative to *R. officinalis* ($p < 0.01$), indicative of anxiolytic neural states.

(3) Under "neutral" conditions, *M. piperita* demonstrated widespread frontocortical, temporal, and occipital β-band facilitation superior to *C. sinensis* and *R. officinalis* ($p < 0.05$). Conversely, under "warm" conditions, thermal comfort discriminated significantly among odorant conditions, with *R. officinalis* eliciting inferior comfort relative to *C. sinensis* ($p = 0.02$).

(4) Under "neutral" conditions, *R. officinalis* or *M. piperita* delivery was recommended to sustain alertness. Under "slightly warm" conditions, *M. piperita* was contraindicated; R. officinalis or *C. sinensis* may be deployed via personalized diffusion systems contingent upon task demands. Under "warm" conditions, strict acoustic control was imperative; in low-noise microenvironments, *M. piperita* integrated with evaporative cooling and ventilation systems attenuated thermal sensation.

CRediT authorship contribution statement

Zi Li: Writing – original draft, Investigation, Data curation. **Xinyue Zhang:** Investigation, Data curation. **Zijing Peng:** Investigation, Data curation. **Ran Xing:** Investigation, Data curation. **Ang Li:** Investigation, Data curation. **Bo Hong:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.buildenv.2026.114576.

Data availability

Data will be made available on request.

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